WORKED SOLUTION

STEP II 1989, Question 12

Mechanics • Equilibrium

STEP II 1989, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q12

The key idea

Only three forces act on the lamina: its weight and the two peg reactions. Smooth pegs push at right angles to the sides they touch, and in a rectangle that means each reaction runs *parallel to the other side*. A body held in equilibrium by three non-parallel forces needs their lines of action to meet at one point. So the point where the two reactions cross must lie on the vertical through the centre. Both results follow from that one fact together with the condition that the pegs are level.

Solution

Setting up

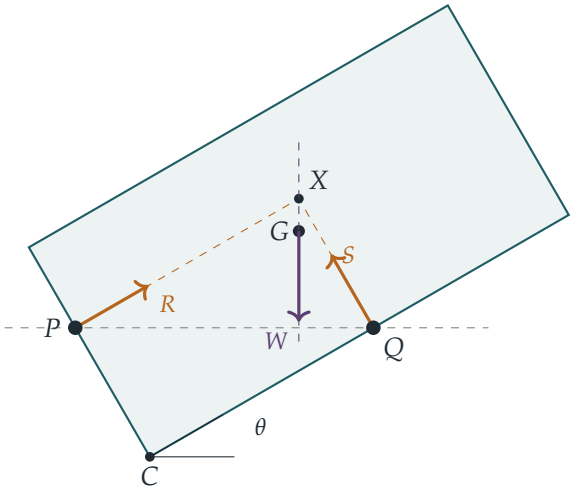
We model the lamina as a rigid uniform rectangle of weight W, with centre G, and the pegs as smooth points P and Q at the same height. Let C be the lowest corner. The side of length $2a$ through C rests on P, with $CP = p$, and the side of length $2b$ through C rests on Q, with $CQ = q$. The $2b$ side makes the acute angle θ with the horizontal, so the $2a$ side (perpendicular to it) makes the angle θ with the vertical.

It is easiest to use coordinates along the sides: from C, measure u along the $2b$ side and v along the $2a$ side. Then

$$C = (0,0), \qquad P = (0,p), \qquad Q = (q,0), \qquad G = (b,a).$$

The point (u,v) has horizontal and vertical displacement from C

$$x = u\cos\theta - v\sin\theta, \qquad y = u\sin\theta + v\cos\theta.$$



The reactions meet on the vertical through G

The reaction R at P is perpendicular to the $2a$ side, so it is parallel to the $2b$ side: its line of action is $v = p$. Similarly the reaction S at Q acts along $u = q$. These lines meet at $X = (q, p)$, the fourth corner of the rectangle $CPXQ$.

Taking moments about X, the reactions have zero moment, so the weight must too: the vertical through G passes through X. In other words, X and G have the same horizontal coordinate:

$$q\cos\theta - p\sin\theta = b\cos\theta - a\sin\theta, \qquad \text{that is,} \qquad (q-b)\cos\theta = (p-a)\sin\theta. \qquad (1)$$

(Resolving gives $R = W\sin\theta$ and $S = W\cos\theta$, both positive, so the pegs really can supply these forces.)

The first result

Since $0 < \theta < \frac{\pi}{2}$, both $\cos\theta$ and $\sin\theta$ are positive, so by (1) $q - b$ and $p - a$ have the same sign. If $p < a$, the right-hand side of (1) is negative, so $q - b < 0$, that is, $q < b$. That is the required result: if the corner is less than a from the peg on the $2a$ side, it is less than b from the other peg.

Ask yourself Why isn't that the end of the problem? Where does d come in?

Equation (1) uses only the moment condition. It is not yet about d, and p and q are not independent: the pegs are level, and they are d apart. We now bring in those two facts.

The relation with d

The pegs are at the same height, so the y-coordinates of P and Q agree:

$$p\cos\theta = q\sin\theta, \qquad \text{so} \qquad p = q\tan\theta. \qquad (2)$$

Their horizontal separation is d:

$$d = x_Q - x_P = q\cos\theta + p\sin\theta = q\cos\theta + \frac{q\sin^2\theta}{\cos\theta} = \frac{q}{\cos\theta}. \tag{3}$$

(Equivalently, CPQ is right-angled at C with hypotenuse $PQ = d$ horizontal, so $q = d\cos\theta$ and $p = d\sin\theta$.)

Now substitute $q = d\cos\theta$ and $p = d\sin\theta$ into the left-hand side of the first form of (1):

$$\begin{aligned} b\cos\theta - a\sin\theta &= q\cos\theta - p\sin\theta \\ &= d\cos^2\theta - d\sin^2\theta \\ &= d\cos 2\theta, \end{aligned}$$

which is the required result.

Checks

- Square case $a = b$, $\theta = \frac{\pi}{4}$: both sides are zero. By symmetry the diagonal CG is vertical, so this is consistent, and it holds for every d, as it should when the square is placed symmetrically.
- With $a = 1$, $b = 2$, $\theta = \frac{\pi}{6}$, equilibrium needs $q = d\cos\theta$ and $d = (b\cos\theta - a\sin\theta)/\cos 2\theta \approx 2.4641$. This gives $p \approx 1.2321 > a$ and $q \approx 2.1340 > b$, consistent with (1). Taking moments about G with $R = W\sin\theta$ and $S = W\cos\theta$ gives zero.
- Numerically, more than 10^5 random equilibria (random a, b, θ with the pegs on the sides) all satisfy the identity to within 10^{-14} and never violate "$p < a \Rightarrow q < b$".

Common slips

- Taking the reaction at a peg to be vertical. A smooth contact pushes perpendicular to the *surface* (here, the side of the lamina), not perpendicular to the ground.
- Mixing up which side makes angle θ with the horizontal: the $2a$ side makes θ with the *vertical*.

Beyond STEP

The three-force principle is a special case of the general statics of rigid bodies: a force system is in equilibrium exactly when its resultant force and its resultant moment about any one point both vanish. At university this becomes the language of wrenches and of virtual work. The virtual-work approach would find this equilibrium by minimising the height of G subject to the lamina keeping contact with the pegs. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully.

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